Retaining Intrinsics in Camera-Tree Schur Complements

A technical description of the GTSFM implementation

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Abstract

Independently reconstructed image clusters can be merged by combining frozen, landmark-free camera likelihoods. To refine calibration during merging, the reduced state must contain both poses and intrinsics: only landmark increments are eliminated. This note derives the retained quadratic, its gauge treatment, and its conversion into similarity-aware factors for a single global solve. It also specifies the public data contract, fixed-intrinsics compatibility mode, and the limits of the resulting approximation.

1 State, observations, and the intended approximation

A leaf reconstruction supplies camera poses, intrinsics, image measurements, and 3D landmarks in an arbitrary local frame. Distinct leaves can contain the same image, identified by filename, but use unrelated local keys and coordinate frames. The merging problem introduces one global pose and calibration for each image and one similarity transformation for each leaf.

Symbol	Meaning
ℓ, i, j	Leaf, globally identified image, and leaf-local landmark indices.
$\bar{T}_{\ell i} \in \mathrm{SE}(3)$	Frozen camera-to-leaf pose estimate.
$\bar{\kappa}_{\ell i} \in \mathbb{R}^3$	Frozen `Cal3Bundler` parameters (f, k_1, k_2).
$\bar{p}_{\ell j} \in \mathbb{R}^3$	Frozen landmark position in the leaf frame.
T_i, κ_i	Global camera pose and per-image calibration to optimize.
$S_\ell \in \mathrm{Sim}(3)$	Similarity mapping global coordinates into leaf ℓ.

The observation model uses one focal length and two radial-distortion coefficients. For $q = T^{-1}p$, let $x = q_x/q_z$, $y = q_y/q_z$, and $r^2 = x^2 + y^2$. Then

$$\pi(T, \kappa, p) = \begin{bmatrix} f(1 + k_1 r^2 + k_2 r^4)x + u_0 \\ f(1 + k_1 r^2 + k_2 r^4)y + v_0 \end{bmatrix}. \tag{1}$$

The principal point (u_0, v_0) is fixed. Occurrences of a shared image must agree on it when intrinsics are variables. Different images have independent calibrations; this API does not tie parameters by physical camera identity.

The producer linearizes each retained observation once at the supplied leaf estimates, with isotropic pixel noise $\sigma^2 I_2$ (default $\sigma = 1$). It does not run bundle adjustment first. Tracks with fewer than two distinct retained cameras are excluded. The consumer subsequently optimizes this frozen camera approximation, without reading landmarks or re-evaluating reprojections. This reconstruction-based producer rebuilds a measurement model; it does not recover the original BA graph, robust weighting, priors, or gauge from saved geometry.

2 Eliminating landmarks while retaining intrinsics

For a leaf, collect pose increments $\delta x \in \mathbb{R}^{6n}$, calibration increments $\delta\kappa \in \mathbb{R}^{3n}$, and landmark increments $\delta p \in \mathbb{R}^{3m}$. Whitened linearization gives the local least-squares problem

$$\min_{\delta x,\delta\kappa,\delta p} \frac{1}{2} \|A_x\delta x + A_\kappa\delta\kappa + A_p\delta p - b\|^2, \qquad b = -r(\bar{T}, \bar{\kappa}, \bar{p}). \tag{2}$$

The residual offset b is retained even when the supplied reconstruction is not at a stationary point. Set $\delta c = [\delta x^\top \ \delta\kappa^\top]^\top$ and $A_c = [A_x \ A_\kappa]$. The normal equations have block structure

$$\begin{bmatrix} H_{cc} & H_{cp} \\ H_{pc} & H_{pp} \end{bmatrix} \begin{bmatrix} \delta c \\ \delta p \end{bmatrix} = \begin{bmatrix} g_c \\ g_p \end{bmatrix}, \qquad H = A^\top A, \quad g = A^\top b. \tag{3}$$

If H_{pp} is nonsingular, profiling out the points yields

$$H_{\text{red}} = H_{cc} - H_{cp}H_{pp}^{-1}H_{pc}, \tag{4}$$

$$g_{\text{red}} = g_c - H_{cp}H_{pp}^{-1}g_p. \tag{5}$$

The pose–intrinsics block is therefore

$$(H_{\text{red}})_{x\kappa} = H_{x\kappa} - H_{xp}H_{pp}^{-1}H_{p\kappa}. \tag{6}$$

Preserving this block allows calibration changes to influence the pose solution and preserves correlations between images induced by shared landmarks.

2.1 Square-root reduction used in the code

The implementation uses QR elimination rather than explicitly forming Equation (4). Conceptually, an orthogonal transformation gives

$$Q^\top \begin{bmatrix} A_p & A_c & b \end{bmatrix} = \begin{bmatrix} R_p & B_p & d_p \\ 0 & B & d \end{bmatrix}. \tag{7}$$

When the landmark columns are eliminable, solving the first block for δp leaves the reduced cost

$$E_{\text{red}}(\delta c) = \tfrac{1}{2}\|B\delta c - d\|^2, \qquad B^\top B = H_{\text{red}}, \quad B^\top d = g_{\text{red}}. \tag{8}$$

Any additive constant discarded by the factor representation does not affect optimization. GTSAM's `GaussianFactorGraph.marginal(retained_keys, EliminateQR)` retains the pose and calibration keys and eliminates only landmark keys. The returned producer summary is ungauged.

2.2 Fixing, retaining, and marginalizing calibration differ

Fixing intrinsics imposes $\delta\kappa = 0$ before reduction, producing a $6n$-dimensional camera system. Retaining intrinsics produces the $9n$-dimensional system above. Marginalizing intrinsics would instead minimize over $\delta\kappa$ and discard their variables; that is not the new behavior. Likewise, an information matrix alone cannot reconstruct the reduced likelihood: g_{red} is essential when $b \neq 0$. For example, x^2 and $(x-3)^2$ have the same quadratic coefficient but different optima.

3 Gauge constraints and leaf Bayes trees

An observable monocular leaf has seven similarity gauge freedoms, even when its intrinsics are retained. For n cameras, the expected ungauged rank is $9n - 7$ in variable-intrinsics mode and $6n - 7$ in fixed-intrinsics mode. Additional self-calibration or geometric ambiguities are rejected; the code does not add implicit calibration priors to make such a system invertible.

The consumer fixes a local coordinate chart before factoring an ungauged leaf. It adds six exact rows fixing the first pose at its frozen estimate and one exact row fixing the baseline length to the farthest camera, linearized at those estimates. A nonzero baseline is required. These constraints act only on poses, not on calibration variables. GTSAM's `NonlinearEqualityPose3` and constrained `RangeFactorPose3` supply the rows.

The augmented graph is factored by multifrontal QR with GTSAM's default ordering. Its Gaussian Bayes-tree conditionals jointly represent the constrained camera quadratic. A conditional can contain any mixture of pose and calibration blocks; block widths are six and three, respectively. Conditionals preserve their linear terms and noise models, including exact constraints.

External producers may supply already gauged graphs with `gauge_fixed=True`. These must be full rank, and the consumer does not add a second coordinate gauge. Mixed gauged and ungauged leaves are supported. Rank checks normalize Jacobian columns and tolerate numerical perturbations of gauge null modes; rank alone cannot certify that an external producer's constraints are purely coordinate choices. That remains part of the producer's contract.

4 Lifting the leaf quadratic to global variables

A similarity variable lets each leaf's constrained local chart move within the global reconstruction. Let $\Phi(S, T)$ denote GTSAM's `Similarity3.transformFrom(Pose3)`. With GTSAM's stored parameters $S = (R_s, t_s, s)$, its action is

$$p \longmapsto s(R_s p + t_s), \qquad \Phi(S, (R, t)) = (R_s R,\ s(R_s t + t_s)). \tag{9}$$

The translation parameter is inside the scale in this convention. The code uses the library operation and its derivatives, rather than implementing an alternate similarity convention.

For each image in leaf ℓ, define frozen-chart increments

$$\xi_{\ell i}(T_i, S_\ell) = \mathrm{Local}_{\bar{T}_{\ell i}}(\Phi(S_\ell, T_i)) \in \mathbb{R}^6, \tag{10}$$

$$\eta_{\ell i}(\kappa_i) = \mathrm{Local}_{\bar{\kappa}_{\ell i}}(\kappa_i) = \kappa_i - \bar{\kappa}_{\ell i} \in \mathbb{R}^3. \tag{11}$$

For conditional q, concatenate the appropriate increments in its actual key order into $z_{\ell q}$. The lifted residual is

$$e_{\ell q} = A_{\ell q} z_{\ell q} - b_{\ell q}. \tag{12}$$

Each shared image has one global calibration but may have different frozen calibration estimates in different leaves. Similarities affect poses only: neither focal length nor distortion coefficients change under a change of world frame or world scale. The residual is passed unwhitened to a `CustomFactor` with the conditional's original noise model, so weighting is applied exactly once.

5 Derivatives, root optimization, and the final reduction

The lifted factors use analytic derivatives in GTSAM's local retraction charts. For a pose block, let A_i^x be its six columns in a conditional and define

$$J_i^{\mathrm{loc}} = D_2 \operatorname{Local}_{\bar{T}_{\ell i}}(\Phi(S_\ell, T_i)), \quad J_i^S = D_S \Phi(S_\ell, T_i), \quad J_i^T = D_T \Phi(S_\ell, T_i). \tag{13}$$

The chain rule gives

$$\frac{\partial e_{\ell q}}{\partial \delta T_i} = A_i^x J_i^{\mathrm{loc}} J_i^T, \tag{14}$$

$$\frac{\partial e_{\ell q}}{\partial \delta S_\ell} = \sum_{i \text{ in pose blocks}} A_i^x J_i^{\mathrm{loc}} J_i^S, \tag{15}$$

$$\frac{\partial e_{\ell q}}{\partial \delta \kappa_i} = A_i^\kappa. \tag{16}$$

Calibration contributes no similarity derivative. In particular, a calibration-only conditional has an identically zero similarity Jacobian. The implementation computes column offsets from the variable types; it never assumes that every conditional block has six columns. Finite-difference tests check the derivatives at nonidentity poses and nonunit similarity scales.

5.1 Initialization and one global solve

Camera identity is resolved by filename, and sorted filenames determine the output indices. Pose-based similarity alignment initializes a connected graph of overlapping leaves. Each usable overlap link needs at least two shared cameras with distinct centers. Global calibrations are initialized from the first occurrence of each image in leaf traversal order.

The input scene tree assembles the leaf evidence bottom-up, without performing an optimization at internal nodes. The root problem is

$$\min_{\{T_i, \kappa_i, S_\ell\}} \sum_{\ell, q} \tfrac{1}{2} \| e_{\ell q}(T, \kappa, S_\ell) \|^2_{\Sigma_{\ell q}}, \qquad S_0 = I, \tag{17}$$

Here $\|e\|^2_\Sigma = e^\top \Sigma^{-1} e$ for stochastic rows; zero-noise rows are exact constraints. Fixing the first leaf's similarity selects the global frame and scale. Levenberg–Marquardt with multifrontal QR solves this problem once at the root. The local pose and calibration estimates used inside the factors remain frozen throughout optimization. The public leaf-to-global transform is S_ℓ^{-1}.

5.2 Final camera information retains calibration

At the converged solution, the assembled graph is linearized again with respect to its global camera and similarity variables. This relinearizes the lifted factors, not the original image measurements. The final marginalization removes only the similarity increments:

$$E_{\mathrm{final}}(\delta c) = \min_{\delta s} E_{\mathrm{root,lin}}(\delta c, \delta s), \qquad \delta c = (\delta x, \delta \kappa). \tag{18}$$

The resulting Gaussian Bayes tree contains global `X(i)` pose keys and `K(i)` calibration keys. It contains no landmarks or similarities and includes the chosen coordinate gauge. The returned scene has optimized poses and calibrations, image metadata, and zero tracks. Its quadratic describes local information in the frozen approximation, not the uncertainty of a newly optimized full bundle-adjustment problem.

6 Public contract, compatibility, and validation

The reconstruction wrapper and the external-factor route agree when supplied the same reduced likelihoods. The reconstruction producer uses `GeneralSFMFactor2Cal3Bundler` to linearize pose, point, and calibration variables. In fixed-intrinsics mode it uses `GeneralSFMFactorPoseCal3Bundler` instead.

```
# Default: eliminate points; retain and optimize pose + intrinsics.
result = optimize_camera_tree(scene_tree, CameraTreeOptions())

# Compatibility: keep intrinsics fixed before linearization.
fixed = optimize_camera_tree(
    scene_tree, CameraTreeOptions(optimize_intrinsics=False))
```

`CameraLeafFactors` carries the following contract:

- `camera_graph` is the complete reduced Gaussian likelihood, including its linear term and weighting; Jacobian and Hessian factors are accepted.
- `estimated_camera_poses` and `camera_metadata` use the same local pose keys. Metadata contains the filename, frozen calibration, and image shape.
- A nonempty `calibration_keys` maps every pose key to one distinct calibration key, disjoint from the pose keys. An empty mapping means fixed intrinsics and preserves existing constructor usage.
- Graph keys must exactly equal the declared pose and calibration keys. Key symbols do not determine types. Arbitrary keys may be reused across leaves.
- `gauge_fixed` declares whether the seven coordinate freedoms have already been removed. The consumer validates rank accordingly.

The consumer infers the variable layout from the summaries; producer options `optimize_intrinsics` and `measurement_sigma` do not reinterpret or reweight external factors. Entire pose-only trees remain supported. Trees mixing pose-only and variable-intrinsics summaries are rejected. Copies are rekeyed to global `X(i)` and `K(i)` without modifying caller-owned input data. Leaf summaries expose frozen `estimated_calibrations`; output cameras expose the optimized values.

6.1 The original-graph marginal API is currently pose-only

The current PR also includes `compute_camera_marginal_factors(graph, values)` from PR #1187. This route linearizes the actual BA factors, including their gauge, priors, and weights. It eliminates non-camera variables first, then uses METIS ordering and QR to produce compact pose conditionals. Its retained keys must contain `Pose3` values: calibration variables, if present, are marginalized out. It does not implement the joint retention described in Sections 2–5, and `optimize_intrinsics` does not configure this API. Preserving original BA factors and retaining intrinsics together requires an external joint summary satisfying the contract above; extending this pose-only helper is separate work.

7 Validation, assumptions, and limitations

At implementation commit `cae83707`, the optimizer and related tree test suite reported **69 passed and one existing dataset-related skip** with Python 3.12 and GTSAM 4.3.0. Coverage includes explicit point minimization versus the retained quadratic, pose–intrinsics cross terms, conditional factorization, weighted analytic derivatives, calibration reconciliation, and both intrinsics modes on the overlapping 12-camera Lund fixture. It also checks frame invariance, external-input equivalence, reused keys, Hessian factors, immutable inputs, and failures for invalid or unobservable inputs. Local NumPy/Accelerate matrix multiplication warnings persisted; explicit finite-result and equivalence checks passed. These are correctness checks, not accuracy or performance benchmarks.

The main numerical checks target distinct parts of the derivation:

1. **Landmark profiling.** Compare reduced objective differences against explicit least-squares minimization of point increments at nonzero residuals. Check that pose–intrinsics cross terms survive and that factorization preserves the constrained quadratic.
2. **Lifted factors.** Compare pose, calibration, and similarity Jacobians with finite differences, including calibration-only conditionals. Check original weighting against the Gaussian factor's cost.
3. **Merged outputs.** Reconcile independent calibration likelihoods with known optima, verify exported intrinsics and retained keys, and compare reconstruction-wrapper results with external-summary results.
4. **Coordinate and input invariants.** Change leaf frames independently, reuse arbitrary local keys, verify caller inputs remain unchanged, and exercise both existing gauge modes and both intrinsics modes.

The reduction is exact for the chosen linearized least-squares problem, up to discarded constants, but not for the original nonlinear BA objective. The root solver re-evaluates pose-chart and similarity transformations while keeping the measurement-derived quadratic frozen. It neither updates points nor relinearizes image observations. Large changes from the frozen estimates can therefore reduce approximation accuracy, even if the root objective decreases.

Leaf likelihoods must not count the same evidence multiple times. The Lund fixture assigns each source track to exactly one leaf while allowing cameras to overlap; the general API does not deduplicate observations. Shared calibration across different images is also outside its contract. Choosing image identity by filename assumes filenames are unique within a leaf and consistent across leaves.

Observability and coordinate conventions are explicit preconditions, not regularization choices. The usable overlap graph must be connected, its links must provide nondegenerate baselines, and each leaf must have the declared rank. Fixed-intrinsics observability does not guarantee self-calibration observability. Principal points must agree across occurrences of a shared image. The final Bayes tree includes the selected gauge; covariance interpretation must respect that choice and the frozen measurement approximation.

The implementation remains separate from production scene merging. Serialization, distributed execution, and further refinement against the original measurements are outside this note's scope. The reconstruction convenience wrapper warns that it cannot preserve the original BA gauge and noise models. The original-graph helper avoids that loss but currently has the pose-only retention boundary described in Section 6.1.

Implementation references. Optimizer source; numerical tests; original PR #1184; original-graph API PR #1187; intrinsics PR #1188. Joint-retention derivations refer to commit `cae83707`; the original-graph API boundary was checked at `20649a17`. The accompanying repository Markdown guide gives usage examples.